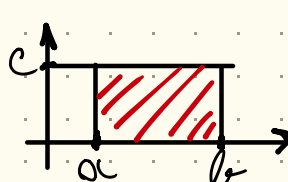


INTEGRALE ORIENTATO

$f: [a, b] \rightarrow \mathbb{R}$ INTEGRABILE, $a < b$

$$\int_a^b f(x) dx =$$


$$\int_a^b c dx =$$


$$= (b - a) \cdot c$$

ANCHE SE $b \leq a$

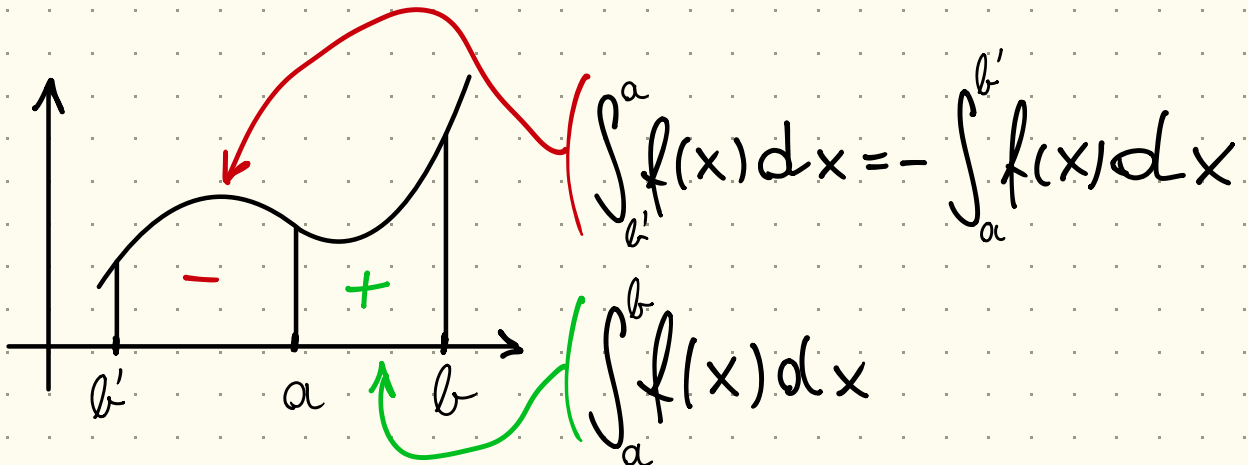
SIA $f: I \rightarrow \mathbb{R}$, I INTERVALLO

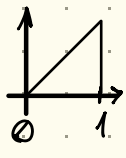
$$\int_a^b f(x) dx = \begin{cases} \text{VISTO SOPRA} & b > a \\ 0 & b = a \\ -\int_b^a f(x) dx & b < a \end{cases}$$

POICHÉ, PER $b < a$

$(b - a) \cdot c = -\text{AREA RETTANGOLO}$

es)



$$\int_1^0 x dx = - \int_0^1 x dx = - \frac{1}{2}$$


• FUNZIONI INTEGRALI

SIA $f: I \rightarrow \mathbb{R}$, I INTERVALLO, T.C. f SIA INTEGRABILE SU OGNI INTERVALLO CHIUSO E LIMITATO $J \subseteq I$

$$F: I \rightarrow \mathbb{R}, \quad F(x) = \int_a^x f(t) dt$$

• LINEARITA'

$$\int_a^b \alpha f(x) + \beta g(x) dx =$$

$$= \alpha \int_a^b f(x) dx + \beta \int_a^b g(x) dx$$

$$\text{es 1)} \int_a^b z f(x) dx = z \int_a^b f(x) dx$$

• ADDITTIVITA' SUI DOMINI

NON CONTA L'ORDINE

$$f: I \rightarrow \mathbb{R}, I \text{ INTERVALLO}, (a, b, c \in I)$$

$$\int_a^b f(x) dx = \underbrace{\int_a^c f(x) dx}_{\text{pink underline}} + \underbrace{\int_c^b f(x) dx}_{\text{red underline}}$$

